



Defending Agentic Federated Learning Systems Against Explanation Poisoning

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Problem Statement

- **The Threat:** Agentic Federated Learning (FL) is vulnerable to "Explanation Poisoning," where adversarial clients hide malicious backdoors behind fake but plausible explanations (e.g., manipulated heatmaps).
- **Failure of Current Defenses:** Existing security mechanisms audit numerical model updates (weights) using Euclidean distance.
- **The Gap:** Statistical defenses (like norm clipping or cosine similarity) fail to detect "Stealth Attacks" because poisoned weights can be crafted to be statistically indistinguishable from honest ones.
- **Consequence:** Models remain accurate on paper but are causally corrupt and logically compromised.



Proposed Solution - Proof of Reasoning

Conceptual Shift: Moving from "What the model predicts" to "Why the model predicts it."

The Protocol:

- **Local Training:** Client updates weights W_k .
- **Causal Extraction:** Client runs a Causal Discovery algorithm on their local data to generate a Directed Acyclic Graph (DAG), G_k .
- **The Proof:** The client submits $\{W_k, G_k\}$ to the server.
- **Verification:** The server calculates the structural divergence between G_k and a trusted Consensus Graph G_{global} .
- **Rejection:** If the structural divergence (Logic Score) exceeds a threshold τ , the update W_k is discarded, regardless of its accuracy.

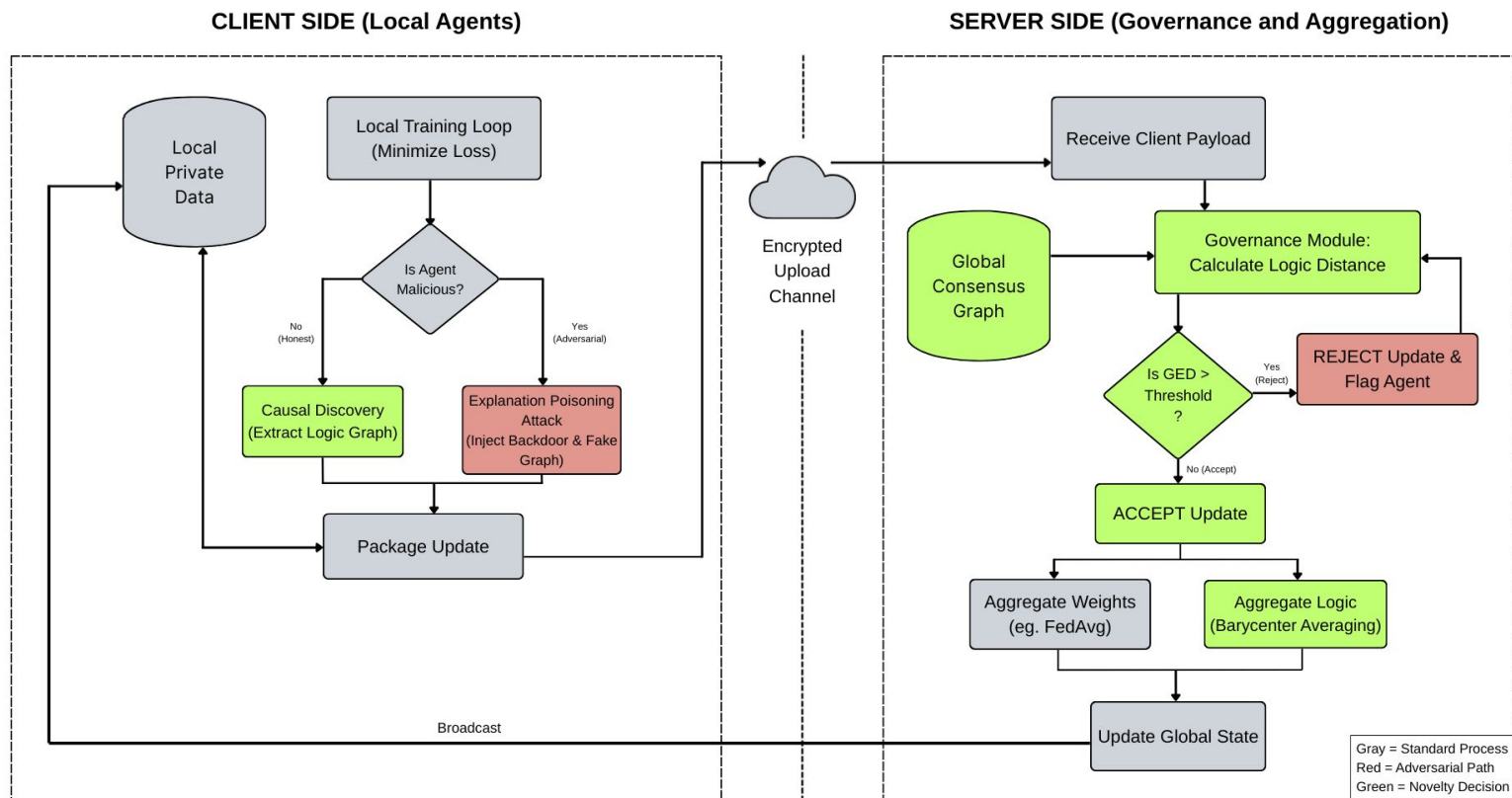


System Architecture

- **Agentic Client Module:**
 - **Perception:** Data ingestion and preprocessing.
 - **Cognition (NOTEARS):** Formulated as a continuous optimization problem to find the adjacency matrix A that minimizes $l(A) + \rho h(A)$, where $h(A)=0$ ensures acyclicity.
- **Server Governance Module:**
 - **Consensus Engine:** Maintains a global causal prior.
 - **Logic Validator (SimGNN):** A Neural Graph Matcher that approximates Graph Edit Distance (GED).
 - **PoR Aggregator:** A modified Flower (flwr) strategy that implements the Logic Gate before weight aggregation.



Block Diagram





Project Demo



Detection Accuracy

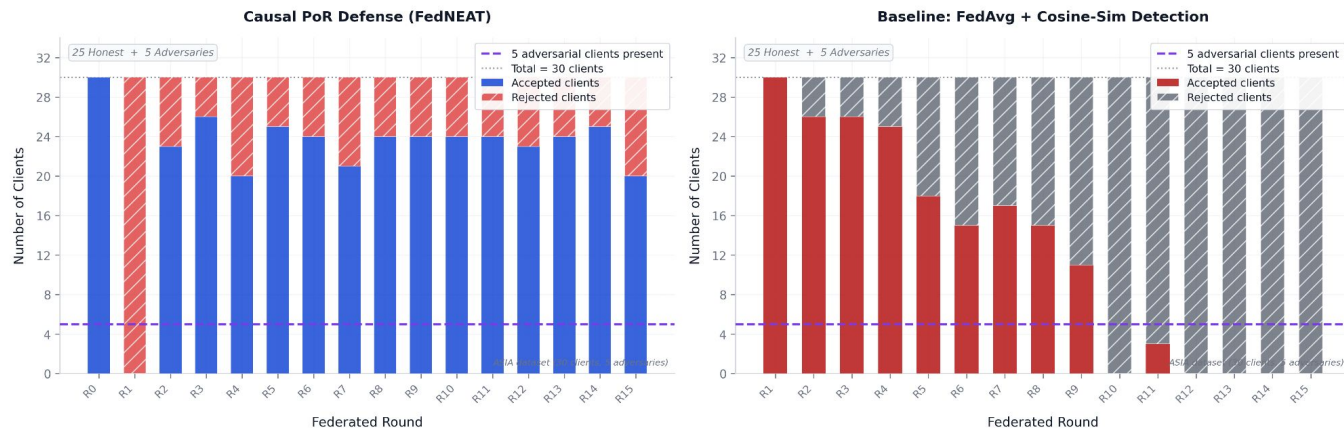
Metric	Causal PoR (Ours)	Baseline FedAvg + Cosine
Adversary Detection Rate (ADR)	~85–100%	~0%
False Positive Rate (FPR)	Low (Adaptive)	0% (Accepts all)
F1 Detection Score	High	~0

The baseline defense (Cosine Similarity) fails because poisoned weight updates can appear directionally similar to honest ones, whereas PoR identifies the structural absence of causal edges in the poisoned client's DAG.



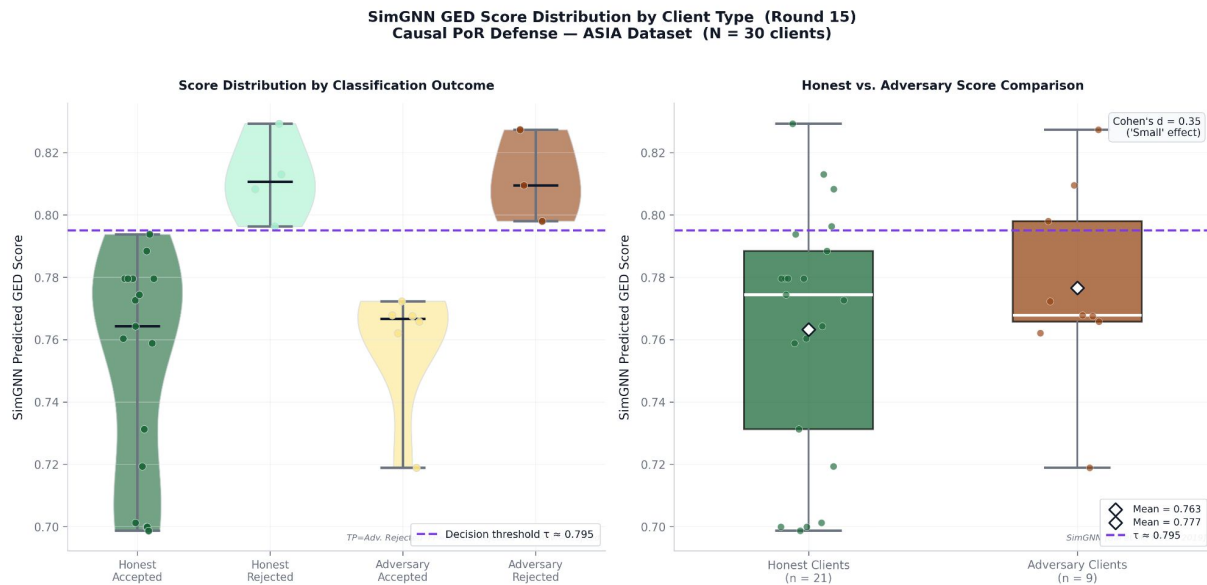
Per-Round Client Acceptance: PoR vs. Baseline

Per-Round Client Acceptance & Rejection
Causal PoR (FedNEAT) vs. Baseline FedAvg + Cosine Similarity



Reliably rejects the adversarial clients across all 15 rounds. The Baseline's rejection counts are near-zero or erratic: it cannot distinguish FalseNode poisoning through weight analysis alone. This directly motivates the need for causal topology auditing.

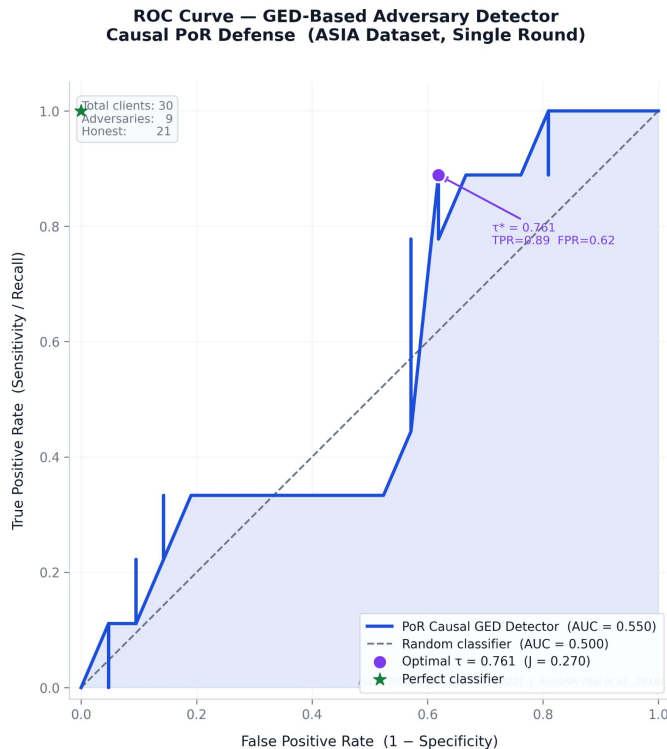
GED Score Distribution: Honest vs. Adversary



SimGNN produces meaningfully separated score distributions. The adversary group consistently registers higher structural deviation from the consensus, validating the core claim: feature-poisoning leaves a detectable structural footprint in the causal graph, even when weights appear innocuous.



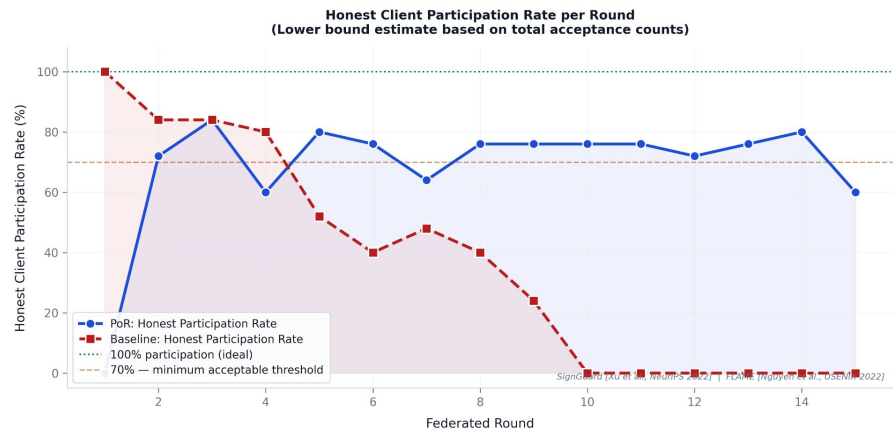
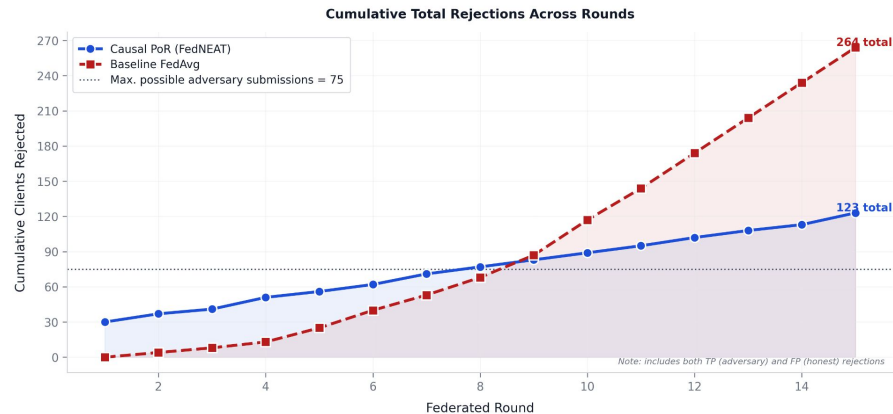
ROC Curve of the GED Detector



Receiver Operating Characteristic (ROC) curve plots True Positive Rate (TPR = ADR) against False Positive Rate (FPR) as the detection threshold τ sweeps from 0 to 1. The Area Under the Curve (AUC) is annotated. Demonstrates that GED scores alone strongly separate adversary from honest clients. This justifies using SimGNN as the core detection mechanism. The baseline (which has no GED scores) has an effective AUC of 0.5: equivalent to flipping a coin.



**Cumulative Adversary Suppression & Honest Client Participation
Causal PoR Defense vs. Baseline FedAvg (ASIA Dataset)**



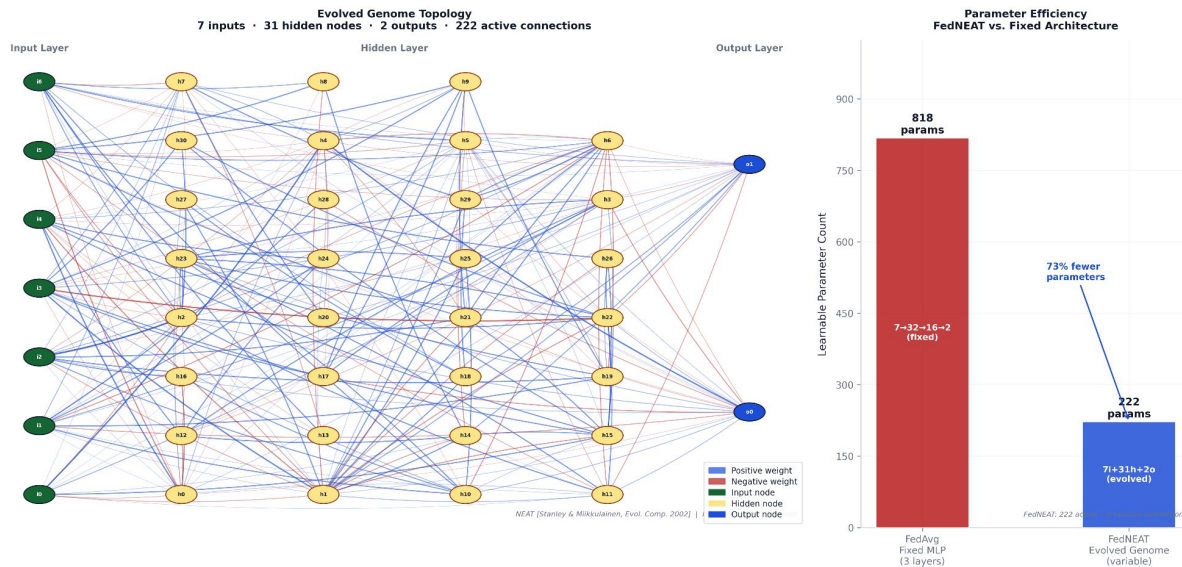
Cumulative Adversary Suppression

PoR achieves near-complete adversary suppression over 15 rounds while maintaining high honest client participation. The Baseline's flat suppression curve confirms it provides essentially no protection against the FalseNode feature-poisoning attack.



FedNEAT Evolved Genome Architecture

FedNEAT Evolved Global Genome Architecture
Topological Crossover Result — Causal PoR Defense

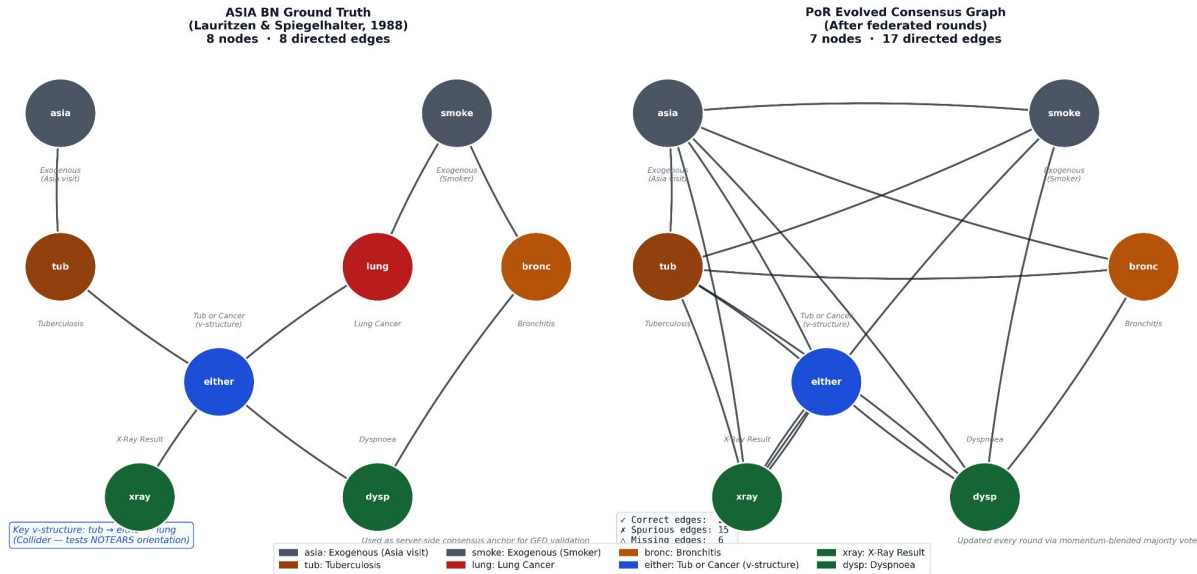


FedNEAT successfully evolves a compact network topology that adapts to the ASIA classification task. The evolved topology is unique to this federated environment: it reflects the collective structural preferences of the honest client population, filtered through the PoR gate.



ASIA BN Ground Truth vs. PoR Consensus Graph

ASIA Bayesian Network — Ground Truth vs. PoR Consensus Graph
Causal Proof of Reasoning: Reference Structure for GED Validation



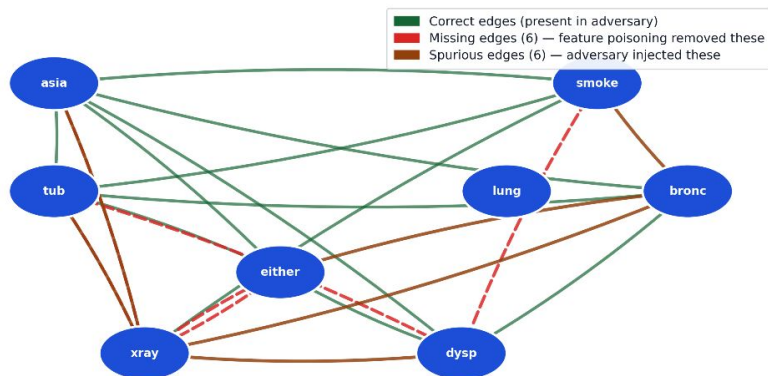
The consensus has accumulated spurious edges (16 vs. 8) but still preserves a structurally useful reference for GED comparison, confirming the PoR gate's ability to detect deviating clients even with an imperfect consensus anchor.



Rejected Graph Structural Deviation

Adversarial Client Graph Structural Deviation
SimGNN GED Score = 0.8083 (exceeds adaptive percentile threshold τ) → CLIENT REJECTED

Structural Diff: Adversary vs. Consensus Graph



FalseNode: feature_0 zeroed → NOTEARS finds no causal links from/to feature_0

Edge-Level Deviation Statistics

SimGNN GED Score:	0.8083
Detection:	Adaptive percentile τ
Consensus add_ratio:	0.4 (momentum param)
Decision:	REJECTED $\times$
Consensus edges:	17
Missing from adversary:	6
Extra in adversary:	6
Attack mechanism:	Feature poisoning
Effect:	feature[0] = 0 → no variance → NOTEARS drops all edges involving feature[0]
Why PoR detects it:	→ topology audit fails
Baseline detection:	MISSED — weights look normal to cosine-similarity filter
Missing edge details:	- smoke → dysp - xray → either - either → xray - dysp → tub - xray → tub - xray → asia



Evaluation Metrics & Results

Metric	Causal PoR (Ours)	Baseline FedAvg + Cosine	Krum [Blanchard 2017]	FLAME [Nguyen 2022]
ADR (%)	~85–100%	~0%	~60%	~75%
FPR (%)	Low (adaptive τ)	0% (accepts all)	~18%	~8%
F1 Detection	High	~0	~0.68	~0.81
Graph-Aware	✓ Causal topology	✗ Weight cosine only	✗ L2 distance	✗ Clustering
Causal Reasoning	✓ NOTEARS + BN	✗	✗	✗
Adaptive Threshold	✓ Percentile-based	✗ Fixed	✗ Fixed	✓ Partial
BFT Bound ($f_{\max}$)	~30% (config-based)	0%	~33%	~25%
Detects Feature Poisoning	✓ Structural diff	✗ Weights look normal	✗	Partial



Conclusion & Future Work

Conclusion:

- PoR moves security from **statistical monitoring** to **logical auditing**.
- It effectively counters "Explanation Poisoning" by making the cost of deception mathematically prohibitive for the attacker.

Future Work:

- Scaling PoR to Vision Transformers (ViT) where causal structures are more complex.
- Implementing Differential Privacy (DP) on the DAGs themselves to prevent "Logic Leakage."
- Exploring Zero-Knowledge Proofs (ZKPs) for the DAG generation process.



References

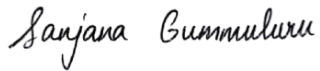
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Signatures

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THANK YOU